Robotics

Degrees Offered

- Graduate certificate in Robotics
- Master of Science in Robotics (non-thesis)
- Master of Science in Robotics (thesis)
- Doctor of Philosophy in Robotics

Program Description

The Robotics program offers a graduate certificate and the degrees of Master of Science and Doctor of Philosophy in Robotics. The graduate certificate is intended for working professionals. The non-thesis MS is designed to prepare candidates for industry careers in robotics. The thesis MS and PhD degrees are designed to prepare students for research careers.

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Combined Program: The Robotics program also offers combined BS+MS degrees. These degrees offer an expedited graduate school application process and allow students to begin graduate coursework while still finishing their undergraduate degree requirements.

Admissions and Program Policies

Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

MS and PhD

The minimum requirements for admission to the MS and PhD degrees in Robotics are:

• Applicants must have a Bachelor's degree, or equivalent, from an accredited institution with a grade-point average of 3.0 or better on a 4.0 scale prior to matriculating into the Robotics degree program.

• Students are expected to have completed the following coursework: 1) two semesters of calculus, 2) differential equations, and 3) data structures. The Robotics graduate admissions committee may require that students who do not meet this expectation demonstrate competency or take remedial coursework. Such coursework may not count toward the graduate degree. The committee will decide whether to recommend regular or provisional admission.

• Graduate Record Examination (Quantitative section) score of 151 or higher (or 650 on the old scale). Applicants who have graduated with a computer science, engineering, or math degree from Mines within the past five years are not required to submit GRE scores.

• Applicants who are not U.S. citizens or permanent residents and whose native language is not English, must prove English proficiency as part of the application process. Refer to the Graduate Admissions website for more details.

• For the PhD program, prior research experience is desired but not required.

Transfer Credit

Graduate-level courses taken at other universities for which a grade equivalent to a B or better was received will be considered for transfer credit with approval of the advisor and/or thesis committee, and home department head, as appropriate. Transfer credits must not have been used as credit toward a bachelor's degree. For the MS degree, no more than nine credits may transfer. For the PhD degree, up to 24 credits of courses may be transferred. In lieu of transfer credit for individual courses, students who enter the PhD program with a thesis-based master's degree from another institution may transfer up to 36 credits in recognition of the course work and research completed for that degree.

Advisor and Thesis Committees

Students must have an advisor from the Robotics faculty to direct and monitor their academic plan, research, and independent studies. Advisors must be full-time permanent members of the faculty. In this context, full-time permanent members of the faculty are those that hold the rank of professor, associate professor, assistant professor, research professor, associate research professor or assistant research professor. Upon approval by the graduate dean, adjunct faculty, teaching faculty, visiting professors, emeritus professors and off-campus representatives may be designated additional co-advisors. A list of Robotics faculty by rank is available in the faculty tab of the catalog.

The department of the advisor is the student's home department.

Master of Science (thesis option) students in Robotics must have at least three members on their thesis committee. In addition to the institutional requirements, at least one committee member who is not the advisor must be Robotics faculty.

Robotics PhD thesis committees must have at least four members. In addition to the institutional requirements, at least one committee member who is not the advisor must be Robotics faculty.

Program Requirements

Graduate Certificate

The graduate certificate will require 12 credits of coursework. The table below summarizes the requirements for the graduate certificate.

Robotics Core	Four courses, one from each focus area's core course list	12.0
Total Semester Hrs		**12.0**

MS Degrees

The MS degrees will require 30 credits, with thesis research options substituting for electives.

MS Non-Thesis (MS-NT)

Students must take 30 credits of coursework to complete the degree. The table below summarizes the requirements for the MS-NT degree.

Robotics Core (Breadth)	Four courses, one from each focus area's core course list	12.0
Robotics Electives (Depth)	Two courses from robotics course list	6.0
Technical Electives	Four courses in any participating robotics department (CSCI, EDNS, EENG, MATH, MEGN)	12.0
Total Semester Hrs		**30.0**

MS Thesis

Students must take 21 credits of coursework and 9 credits of MS thesis research to complete the degree. The table below summarizes the requirements for the MS Thesis degree.

At the conclusion of the MS thesis, the student must make a formal presentation and defense of their thesis research. A student must pass this defense to earn an MS degree.

Robotics Core (Breadth)	Four courses, one from each focus area's core course list	12.0
Robotics Electives (Depth)	Two courses from robotics course list	6.0
Technical Electives	One course in any participating robotics department (CSCI, EDNS, EENG, MATH, MEGN)	3.0
ROBO707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	9.0
Total Semester Hrs		**30.0**

PhD Degree

The Robotics PhD requires 36 credits of coursework plus 36 research credits. The table below summarizes the coursework requirements. PhD students must additionally complete a qualifying examination, a thesis proposal, and a thesis defense.

Robotics Core (Breadth)	Four courses, one from each focus area's core course list	12.0
Robotics Electives (Depth)	Four Courses from robotics course list	12.0
Technical Electives	Four courses in any participating robotics department (CSCI, EDNS, EENG, MATH, MEGN)	12.0
ROBO707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	36.0
Total Semester Hrs		**72.0**

Robotics PhD Qualifying Examination

The Robotics PhD Qualifying Examination will test a student's ability to conduct research in their chosen area. The qualifier will have two components: a coursework component and a research component.

Coursework Qualifier: To satisfy the coursework component of the qualifier, the student must complete their four selected Robotics core courses with a minimum grade of B in each class.

Research Qualifier: The research qualifier consists of a research project. Robotics PhD students must take the qualifying examination by the end of their fourth semester (typically by the end of their second year). The examination will be evaluated by a committee consisting of at least the student's advisor, a Robotics-affiliated faculty member, and one additional faculty member.

For the qualifier, the student will conduct a literature review of 30-40 papers and perform a research project approaching the level necessary for a conference publication. The research project must be approved by the advisor and committee and will likely be some combination of the following:

- Design, analyze and/or simulate a novel robot system,
- Design and evaluate new algorithms or systems for an important research problem,
- Develop a new research software system,
- Solve a set of theoretical problems.

The deliverables will be a literature review (3-4 pages, IEEE style [4]), a research report (4-5 pages, IEEE style), and a research presentation (30 minutes to present, plus questions) delivered to the committee.

At the conclusion of the qualifier presentation, each committee member will vote their evaluation as one of “Strong Support,” “Weak Support,” or “Do not support.” The student must receive at least two “Strong Support” votes to pass. In the case the student does not pass, the committee may decide to offer a “conditional pass” based on extra conditions, such as revisions to the report or additional experiments, the student must perform to pass the qualifier. The committee will set an explicit deadline for student to complete the extra conditions. If the student does not meet the extra conditions as determined by the committee by the deadline, the “conditional pass” becomes a “fail.” If the student does not pass the qualifier on their first attempt (inclusive of a conditional pass), they may make one additional attempt to pass; the same conditional pass procedure may also be applied to the second attempt. A student who fails the qualifier on the second attempt may not continue in the program.

Robotics PhD Proposal and Defense

After passing the qualifying examination, the student must prepare a written thesis proposal and present it formally to the student’s thesis committee and other interested faculty. Typically, the proposal will take place within 24 months of the student completing the qualifier.

The committee for the thesis proposal and defense will follow institutional requirements. Additionally, at least one committee member who is not the advisor must be robotics-affiliated faculty.

At the conclusion of the student’s PhD program, the student must make a formal presentation and defense of their thesis research. A student must “pass” this defense to earn a PhD degree. Typically, the defense will take place within 24 months of the student completing the thesis proposal.

Robotics Course List

The Robotics courses are divided into four focus areas. Each focus area is comprised of core courses and electives, as detailed below. Beyond the courses in these four focus areas, there is also a list of additional Robotics electives.

Perception

Core Courses

ROBO513	ROBOT PROGRAMMING AND PERCEPTION	3.0
ROBO517	INTRODUCTION TO COMPUTER VISION	3.0
ROBO529	ESTIMATION THEORY AND KALMAN FILTERING	3.0
ROBO537	ROBOTIC MAPPING AND LOCALIZATION	3.0

Elective Courses

CSCI508	ADVANCED TOPICS IN PERCEPTION AND COMPUTER VISION	3.0

Cognition

Core Courses

ROBO533	PLANNING FOR SENSING, PERCEPTION, AND COVERAGE	3.0
ROBO534	ROBOT PLANNING AND MANIPULATION	3.0
ROBO535	ADVANCED MACHINE LEARNING	3.0

Elective Courses - None.

Action

Core Courses

ROBO554	ROBOT MECHANICS: KINEMATICS, DYNAMICS, AND CONTROL	
ROBO550	MECHATRONICS	3.0
ROBO565	ADVANCED ROBOT CONTROL	3.0
ROBO567	THEORY AND DESIGN OF ADVANCED CONTROL SYSTEMS	3.0

Elective Courses

EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0
ROBO566	MODERN CONTROL DESIGN	3.0

Interaction & Society

Core Courses

ROBO572	ROBOT ETHICS	3.0
ROBO576	HUMAN-ROBOT INTERACTION	3.0

Elective Courses

CSCI5XX	LINGUISTIC HUMAN-ROBOT INTERACTION	3.0

Additional Robotics Electives

CSCI561	THEORY OF COMPUTATION	3.0
CSCI562	APPLIED ALGORITHMS AND DATA STRUCTURES	3.0
CSCI565	DISTRIBUTED SYSTEMS	3.0
CSCI572	COMPUTER NETWORKS II	3.0
EENG511	CONVEX OPTIMIZATION AND ITS ENGINEERING APPLICATIONS	3.0
EENG521	NUMERICAL OPTIMIZATION	3.0
MEGN586	LINEAR OPTIMIZATION	3.0
MEGN587	NONLINEAR OPTIMIZATION	3.0
MEGN588	INTEGER OPTIMIZATION	3.0
MEGN686	ADVANCED LINEAR OPTIMIZATION	
MEGN688	ADVANCED INTEGER OPTIMIZATION	3.0
ROBO509	SPACE ROBOTICS	3.0

ROBO509. SPACE ROBOTICS. 3.0 Semester Hrs.
This online course introduces students to the design, analysis, and operation of robotic spaceflight missions. By integrating space exploration with robotics, students will explore planetary science questions, develop mission plans, and evaluate the technologies required to achieve mission objectives. Topics include sensor systems, automation trade-offs, and rover mobility. The course prepares students for careers in aerospace and robotics by providing essential technical, analytical, and organizational skills to solve complex challenges in these fields. Prerequisites: SPRS501; Desired, some background in either aerospace engineering, mechanical engineering, electrical engineering, computer science, or planetary science.
Course Learning Outcomes

- Evaluate state of the art planetary science questions and technology goals from authoritative space exploration roadmaps to inform the development of new robotic missions.
- Analyze the progression of spaceflight technologies, focusing on the advances in rover mobility and sensing capabilities for planetary surface exploration.
- Design an extraplanetary surface robotic mission with a science or technology goal derived from roadmaps.
- Assess the data signals from sensors necessary to produce the end data product, which is informed by the science or technology goal.
- Plan the mission operations and robot actuators necessary to achieve mission objectives.
- Differentiate operations between robot automation and human teleoperation along the metrics of risk, SWaP, complexity, and timeline.
- Develop a perception framework and path planning strategy informed by mission requirements and a concept of operation.
- Design basic mobility algorithms constrained by robot capabilities and limitations in Moon environment.

ROBO513. ROBOT PROGRAMMING AND PERCEPTION. 3.0 Semester Hrs.
In this class students will learn the basics of integrated robot system programming and the design and use of algorithms for robot perception. Students will learn how to use the ROS robot middleware for the design of robot systems for perceiving and navigating the world; develop reinforcement learning based models for perception-informed autonomous navigation; and develop computational models for 3D robot perception and perceptual representation of human data.
Course Learning Outcomes

- 1. Explain the basic concepts in human-centered robotics
- 2. Model and analyze human behaviors for human-robot interaction applications
- 3. Recognize the cutting-edge human-centered robotics research and applications
- 4. Apply the learned knowledge to other fields

ROBO517. INTRODUCTION TO COMPUTER VISION. 3.0 Semester Hrs.
(I) Computer vision is the process of using computers to acquire images, transform images, and extract symbolic descriptions from images. This course provides an introduction to this field, covering topics in image formation, feature extraction, location estimation, and object recognition. Design ability and hands-on projects will be emphasized, using popular software tools. The course will be of interest both to those who want to learn more about the subject and to those who just want to use computer imaging techniques. 3 hours lecture; 3 semester hours. Prerequisite: Undergraduate level knowledge of linear algebra, statistics, and a programming language.
Course Learning Outcomes

- 1. Be able to analyze and predict the behavior of image formation, transformation, and recognition algorithms
- 2. Be able to design, develop, and evaluate algorithms for specific applications
- 3. Be able to use software tools to implement computer vision algorithms
- 4. Communicate (in oral and written form) methods and results to a technical audience

ROBO529. ESTIMATION THEORY AND KALMAN FILTERING. 3.0 Semester Hrs.
Estimation theory considers the extraction of useful information from raw sensor measurements in the presence of signal uncertainty. Common applications include navigation, localization and mapping, but applications can be found in all fields where measurements are used. Mathematic descriptions of random signals and the response of linear systems are presented. The discrete-time Kalman Filter is introduced, and conditions for optimality are described. Implementation issues, performance prediction, and filter divergence are discussed. Adaptive estimation and nonlinear estimation are also covered. Contemporary applications will be utilized throughout the course. Offered Spring semester of odd years. 1.5 hours lecture; 1.5 hours other; 3 semester hours.
Course Learning Outcomes

- Use Bayes' rule to calculate a statistical inference. Given a description of a stochastic process, calculate the joint and conditional probabilities for this process.
- Using the appropriate algorithm, calculate the probability distribution function for the state of a dynamic system with stochastic inputs.
- " Build a model of a dynamic system that includes a probabilistic description of uncertain inputs.
- Design and implement an algorithm to estimate the internal states of a linear system with input signals that are Gaussian stochastic processes.
- Design and implement an algorithm to estimate the internal states of general systems with general stochastic inputs.

ROBO533. PLANNING FOR SENSING, PERCEPTION, AND COVERAGE. 3.0 Semester Hrs.
Identification of applications of theory. Prerequisites: Graduate standing; Recommended: CSCI 406 (Algorithms); CSCI 358 (Discrete Mathematics).
Course Learning Outcomes

- Formulate relevant optimization problems.
- Analyze optimization problems.
- Prove properties of planning algorithms.
- Design and implementation of relevant methods.
- Research reading and communication.
- Design and solution for perception planning and sensing.
- Identification of applications of theory.
- Communication of research works.
- Breadth of applications.

ROBO534. ROBOT PLANNING AND MANIPULATION. 3.0 Semester Hrs.
An introduction to planning in the context of robotics covering symbolic and motion planning approaches. Symbolic computation, symbolic domains, and efficient algorithms for symbolic planning; Robot kinematics, configuration spaces, and algorithms for motion planning. Applications of planning will focus on manipulation problems using robot arms.
Course Learning Outcomes

- 1 - Implement algorithms for symbolic computation
- 2 - Construct symbolic planning domains for new scenarios
- 3 - Implement algorithms for symbolic planning via constraint-solving and heuristic search
- 4 - Implement algorithms for sampling-based motion planning
- 5 - Construct kinematic models of robot manipulators
- 6 - Analyze planning algorithms for key properties: correctness, completeness, optimality
- 7 - Evaluate the suitability of different planning approaches and apply appropriate algorithms to new planning scenarios
- 8 - Communicate implementations, analysis, and evaluation in written and oral form

ROBO535. ADVANCED MACHINE LEARNING. 3.0 Semester Hrs.
Machine learning is the study of computer algorithms that improve automatically through experience. Machine learning systems do not have to be programmed by humans to solve a problem; instead, they essentially program themselves based on examples of how they should behave, or based on trial and error experience trying to solve the problem. This course aims at provide students with an understanding of the capabilities of machine learning (especially for deep learning due to its state-of-the-art performance for predicting and understanding data), and the knowledge to formulate the real-world problem to solve it effectively by a combination of computational idea motivations, learning theories, mathematical and optimization backgrounds/tools.

ROBO537. ROBOTIC MAPPING AND LOCALIZATION. 3.0 Semester Hrs.
This course introduces the principles and techniques of simultaneous localization and mapping (SLAM), a core capability in modern robotics. Students will learn how to leverage robot sensors (including cameras and laser scanners) to construct 3D maps of an environment and estimate the robot's position within it. The course covers both the front-end techniques, which process raw sensor data (e.g., camera images) to generate initial 3D maps and position estimates, and the back-end techniques, which refine these outputs by filtering outliers, optimizing trajectories, and fusing local maps into a consistent global representation. The course includes hands-on homework and projects in which students will implement SLAM algorithms and test them on real-world data collected by robots. Prerequisites: Familiarity with: - Linear algebra, calculus, probability, and statistics - Linux, Algorithms, C++, and Python - Computer vision.
Course Learning Outcomes

- Learn foundational mathematical tools (from linear algebra, optimization, and statistics) and use them to formulate, analyze, and solve different problems in robotic mapping and localization.
- Develop a deep understanding of front-end SLAM techniques, including feature extraction, pinhole camera modeling, and multi-view geometry, to generate initial 3D maps and pose estimates.
- Develop a deep understanding of back-end SLAM techniques, including graph optimization, outlier rejection, and loop closure, to refine local maps into consistent global representations.
- Integrate state-of-the-art SLAM algorithms in Python or C++, learn how to use them in the Robot Operating System (ROS) framework, and experiment with running them on real-world sensor data to build 3D environment maps.
- Learn standard benchmarking metrics used to evaluate the performance of SLAM systems, and implement them to critically assess the accuracy, efficiency, and robustness of these systems.
- Build a complete SLAM pipeline from scratch in a team-based project, and develop an understanding of the required real-world trade-offs between performance, runtime, and robustness.
- Identify existing challenges and open problems in SLAM, and propose innovative research directions to advance the state of the field.

ROBO550. MECHATRONICS. 3.0 Semester Hrs.
A course focusing on implementation aspects of mechatronic and control systems. Significant lab component involving embedded C programming on a mechatronics teaching platform, called a haptic paddle, a single degree-of-freedom force-feedback joystick.
Course Learning Outcomes

- 1. Become proficient in mechanical system modeling, system identification and simulations.
- 2. Develop an understanding of how control theory is applied and implemented in practice.
- 3. Learn fundamentals of and how to use semiconductor devices in mechatronic systems
- 4. Learn the basics of sensor and actuator theory, design, and application
- 5. Gain experience in embedded C programming for mechatronic systems
- 6. Gain experience in research article reading and technical presentations

ROBO554. ROBOT MECHANICS: KINEMATICS, DYNAMICS, AND CONTROL. 3.0 Semester Hrs.
Mathematical representation of robot structures. Mechanical analysis including kinematics, dynamics, and design of robot manipulators. Representations for trajectories and path planning for robots. Fundamentals of robot control including, linear, nonlinear and force control methods. Introduction to off-line programming techniques and simulation. Robotics students should register for ROBO554, not MEGN544. 3 hours lecture; 3 semester hours. Prerequisites: EENG307 and MEGN441.

ROBO565. ADVANCED ROBOT CONTROL. 3.0 Semester Hrs.
The goal of this course is to give the students an introduction to a fundamental working knowledge of the main techniques of intelligent learning-based control and their applications in robotics and autonomous systems. Specific topics include neural network based control, model predictive control, reinforcement learning based control, fuzzy logic control, and human-in-the-loop control.

ROBO566. MODERN CONTROL DESIGN. 3.0 Semester Hrs.
Control system design with an emphasis on observer-based methods, from initial open-loop experiments to final implementation. The course begins with an overview of feedback control design technique from the frequency domain perspective, including sensitivity and fundamental limitations. State space realization theory is introduced, and system identification methods for parameter estimation are introduced. Computer based methods for control system design are presented. Prerequisites: EENG307.
Course Learning Outcomes

- Model and analyze single-input single-output (SISO) systems using both transfer function and state space realizations in continuous- and discrete-time.
- Design and test controllers for these systems.